

ORIGINAL RESEARCH

Na/K-ATPase Signaling in Adipocytes Promotes Atherosclerosis

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BACKGROUND: Adipocyte dysfunction is closely associated with oxidative stress and chronic inflammation, which contribute to systemic metabolic disturbances and atherosclerosis. We previously identified the NKA (Na/K-ATPase) $\alpha 1$ subunit as a signal transducer that activates Src-family kinases and promotes oxidative stress and inflammation in various cell types, including adipocytes and macrophages. NaKtide, a peptide inhibitor of NKA signaling, has been shown to reduce systemic oxidative stress and inflammation in vivo. In this study, we investigated the role of adipocyte-specific NKA signaling in atherosclerosis.

METHODS: Adipocyte-specific NaKtide was delivered to *Apoe*^{-/-} mice using a lentiviral vector under the adiponectin promoter. The mice were then fed a Western diet for 12 weeks to induce atherosclerosis and then assessed for atherosclerotic plaque burden in the aortic arch and at the level of the aortic sinus. Inflammatory and oxidative stress markers were analyzed in adipose tissue and plasma.

RESULTS: Adipocyte-specific NaKtide reduced atherosclerotic plaque area by 67% in the aortic arch and by 48% in the aortic sinus. CD68⁺ (cluster of differentiation 68) macrophage content and α -SMA⁺ (α -smooth muscle actin) smooth muscle cell content in the aortic sinus were decreased by 45% and 53%, respectively. These vascular improvements were accompanied by dampened adipose tissue inflammation and oxidative stress, improved glucose tolerance, and reduced systemic inflammation.

CONCLUSIONS: These findings highlight a critical contributing role for adipocyte NKA signaling in atherosclerosis, suggesting an important endocrine and paracrine influence of adipose tissue on large artery atherogenesis and supporting the therapeutic potential of targeting NKA in cardiometabolic disease.

Key Words: adipocytes ■ atherosclerosis ■ inflammation ■ oxidative stress ■ Src-family kinases

Atherosclerosis is a chronic inflammatory condition, driven by lipid accumulation in the arterial wall, systemic metabolic dysfunction, and oxidative stress.¹⁻³ Among the metabolic organs contributing to this pathological state, adipose tissue plays a pivotal role.^{4,5} Under conditions such as obesity or high-fat feeding, adipocytes undergo functional changes that result in increased secretion of proinflammatory cytokines and reactive oxygen species, which in turn exacerbate systemic inflammation and promote vascular injury.^{6,7}

The NKA (Na/K-ATPase) $\alpha 1$ subunit, beyond its canonical cation-transport function, was identified nearly 2 decades ago as a signal transducer through its ability to interact directly with SFKs (Src-family kinases).⁸⁻¹¹ The cytoplasmic domain of NKA $\alpha 1$ sequesters SFKs at the inner leaflet of the plasma membrane, thereby holding them in an inactive, inaccessible state. In response to extracellular stimuli, conformational changes in NKA release activated SFKs, triggering downstream signaling cascades that promote oxidative stress and inflammation

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Nonstandard Abbreviations and Acronyms

Adipo-NaKtide	adipocyte-specific NaKtide
α-SMA	α -smooth muscle actin
AKT	protein kinase B
CD	cluster of differentiation
CTR	control diet
GFP	green fluorescent protein
IL-6	interleukin-6
NKA	Na/K-ATPase
ORO	Oil Red O
oxLDL	oxidized low-density lipoproteins
pAKT	phosphorylated protein kinase B
pSFK	phosphorylated Src-family kinase
SFK	Src-family kinases
tAKT	total protein kinase B
TNF-α	tumor necrosis factor- α
tSKF	total Src-family kinases
VFAT	visceral fat adipose tissue
WD	Western diet

in multiple cell types, including macrophages and adipocytes.^{9,11} A 20-amino acid mimetic peptide based on the sequence of the SFK-binding domain of the NKA α 1 subunit was developed¹² to inhibit the NKA-SFK signaling axis and was termed NaKtide. NaKtide binds the conserved SFK kinase domain, effectively blocking its activation and downstream signaling. Our previous studies showed that systemic administration of pNaKtide (a cell-permeable version of NaKtide) reduced oxidative stress, improved metabolic parameters, and attenuated inflammation in animal models of chronic diseases.^{13–15} However, the specific role of adipocyte NKA signaling in the progression of atherosclerosis has not yet been defined.

We previously demonstrated that NKA α 1 forms a functional complex with CD36 (cluster of differentiation 36)¹⁶ in adipocytes. CD36 is a fatty acid transporter and scavenger receptor known to play an important role in macrophage uptake of oxLDL (oxidized low-density lipoproteins), foam cell formation, and atherosclerosis.¹⁷ The NKA-CD36 signaling axis promotes lipid accumulation, reactive oxygen species generation, proinflammatory cytokine expression, and metabolic reprogramming under atherogenic conditions in various cell types, including adipocytes.^{16,18,19} These proatherogenic effects were abolished by pNaKtide in vitro,¹⁶ suggesting that NKA signaling may contribute to vascular inflammation and atherogenesis. Despite these findings, the in vivo significance of adipocyte-specific NKA signaling in atherosclerosis remains unknown. In this study, we addressed this gap by expressing NaKtide selectively in adipocytes using a lentiviral vector under the control of the adiponectin promoter (adipocyte-specific NaKtide

What Are the Clinical Implications?

The present study identifies adipocyte Na/K-ATPase signaling as a previously unrecognized contributor to atherosclerosis and suggests that targeting adipocyte-derived inflammatory communication may represent a novel therapeutic strategy for cardiovascular disease. Obesity and metabolic syndrome are characterized by adipose tissue inflammation, which is strongly associated with increased cardiovascular risk but remains difficult to target clinically. Our findings demonstrate that inhibition of adipocyte Na/K-ATPase signaling with adipocyte-specific NaKtide attenuates adipose oxidative stress, reduces systemic inflammation, and suppresses macrophage activation and atherosclerotic lesion development. These observations support the concept that dysfunctional adipose tissue is not merely a consequence of cardiometabolic disease but also acts as an active driver of vascular inflammation. Importantly, therapies that target adipocytes may complement current lipid-lowering strategies by addressing residual inflammatory cardiovascular risk, particularly in individuals with obesity, insulin resistance, and metabolic syndrome. Thus, adipocyte Na/K-ATPase signaling represents a promising therapeutic target for preventing obesity-associated cardiovascular disease.

[Adipo-NaKtide]) in high-fat diet-fed *Apoe*^{-/-} mice and found that it reduced atherosclerotic burden in the aortic arch and sinus, dampened adipose tissue inflammation and oxidative stress, improved glucose tolerance, and lowered circulating levels of systemic inflammatory markers. These findings reveal a previously unrecognized role of adipocyte NKA signaling in the pathogenesis of atherosclerosis and suggest that targeting adipocyte NKA could be an effective therapeutic approach to limit metabolic inflammation in cardiovascular disease.

MATERIALS AND METHODS

Data Availability

The data that support the findings of this study are available from the corresponding author on reasonable request. All data are available in the main text and supplemental materials.

Mouse Models and In Vivo Studies

All animal studies were approved by the animal care and use committees of Marshall University and the Medical College of Wisconsin in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals. All mice involved in this study were from a C57BL/6J genetic background. *Apoe* null mice (*Apoe*^{-/-}; 6–8 weeks old, male and female) were purchased from the Jackson Laboratory. On arrival at the Robert C. Byrd Biotechnology Science Center Animal Resource Facility, the mice were housed in designated rooms equipped with cages

that supplied purified air on a 12-hour light/dark cycle. Mice were fed a normal laboratory diet (control diet [CTR]) or a high-fat Western diet (WD) for 3 months. WD is a well-known model of diet-induced oxidative stress and metabolic syndrome, and it contributes to chronic metabolic imbalance and the development of atherosclerosis.^{19,20} The WD was purchased from Envigo (Indianapolis, IN) and contained 42.1% fat, 42.7% carbohydrate, and 15.2% protein, yielding 4.5 kJ/g. The CTR was purchased from PicoLab Laboratory Rodent Diet and contained 13.6% fat, 57.5% carbohydrate, and 28.9% protein.

Lentiviral vectors expressing either GFP (green fluorescent protein)-tagged NaKtide or GFP alone under the control of an adiponectin promoter were constructed by VectorBuilder Inc to yield adipocyte-specific expression, as previously reported.²¹ Lentiviral constructs (100 μ L, 2×10^9 TU/mL) in saline were injected intraperitoneally into *Apoe*^{-/-} mice as described previously.²¹ Body weight was monitored weekly throughout the 12-week experimental period. At the end of the study, after an 8-hour fast, blood samples were collected from the tail vein, and plasma was isolated for measurement of lipids, glucose, and cytokines. Mice were euthanized 12 weeks after the dietary interventions, and at the time of termination, blood, visceral fat (visceral fat adipose tissue [VFAT]), heart, and aorta were collected. Plasma was isolated, flash-frozen in liquid nitrogen, and stored at -80°C until measurement of plasma biomarkers. Mice were euthanized by ketamine overdose (100 mg/kg body weight) and perfused via the left ventricle with 10 mL PBS. After perfusion, the aorta was dissected, cleaned, and fixed in 4% paraformaldehyde. Heart tissues containing the aortic roots and adipose tissues were carefully dissected and embedded in optimal cutting temperature solution for cryosectioning. Some heart, aorta, and adipose tissues were flash-frozen in liquid nitrogen and stored at -80°C for Western blots and quantitative real-time polymerase chain reaction.

Measurement of Protein Carbonylation

Protein carbonylation in VFAT was quantified using a Protein Carbonyl ELISA Assay Kit (BioCell Corporation, Auckland, New Zealand), following the protocol provided by the manufacturer. The assay is based on the derivatization of protein carbonyl groups with 2,4-dinitrophenylhydrazine, forming dinitrophenyl adducts that are specifically detected using an anti-dinitrophenyl antibody. Derivatized proteins were immobilized on ELISA plates and detected using a horseradish peroxidase-conjugated secondary antibody with colorimetric readout at 450 nm. Quantification was performed using oxidized protein standards supplied with the kit and processed in parallel with samples. According to the standard curve provided by the manufacturer, the limit of quantification of the assay is 0.08 nmol carbonyls/mg protein, corresponding to the lowest nonzero standard concentration. All samples were assayed under identical experimental conditions to minimize technical variability.

Estimation of SFK Active Site Phosphorylation [pY419] and Protein Kinase B Active Site Phosphorylation [ser473] by Western Blot analysis

VFAT was homogenized with RIPA buffer, and pSFK (phosphorylated SFK) was detected using a polyclonal anti-SFK

(pY419) phospho-specific antibody (Invitrogen). In addition, pAKT (phosphorylated AKT [protein kinase B]) was detected with a polyclonal anti-AKT (ser473) phospho-specific antibody (Cell Signaling). The same membranes were then stripped and blotted with a monoclonal antibody against tSFK (total SFK; Santa Cruz) and a polyclonal antibody against tAKT (total AKT; Cell Signaling). Expression of pSFK and AKT was expressed as the ratio of pSFK/tSFK and pAKT/tAKT, respectively, with measurements normalized to control samples.

Measurement of Plasma Cytokines, Leptin, and Glucose Profile

Plasma IL-6 (interleukin-6), TNF- α (tumor necrosis factor- α), and leptin were measured using ELISA kits, according to the protocol provided by the manufacturer (Abcam, Cambridge, MA). Furthermore, the glucose tolerance test was performed using a glucometer. Mice were fasted for 8 hours before testing, following an established protocol.¹³ A 10% glucose solution was administered intraperitoneally at a dose of 2 g/kg body weight, and blood glucose levels were measured from the tail vein at baseline (0 minutes) and at 30, 60, 90, and 120 minutes after injection.



Oil Red O Staining of the Aortic Arch and Aortic Sinus

To quantify the atherosclerotic plaque area, aortic arches were isolated and stained with Oil Red O (ORO). Briefly, aortas were opened longitudinally and rinsed with PBS, followed by staining with a 2.4% ORO solution in isopropanol for 15 minutes. The tissues were then destained in 60% isopropanol until the background was clear. After washing, the samples were stained en face with ORO for 20 minutes, pinned, and examined under a stereomicroscope. The lesion area was quantified as a percentage of the ORO-stained area in the total aortic arch area using ImageJ software.

For aortic sinus staining, fresh-frozen hearts were cut into 6- μ m sections, starting from the aortic valve leaflets and moving toward the ascending aorta. Four serial sections, covering a 300- μ m segment of the aortic root at 72- μ m intervals, were chosen. All sections underwent standard processing and were imaged using a Motic system. Sections were stained with a 2.4% ORO solution for 30 minutes, followed by washing 3x with PBS. The cross-sectional lesion area for each mouse was calculated using the mean value of 3 to 4 series of sections. The lesion area was quantified as a percentage of the ORO-stained area of the total aorta sinus area using ImageJ software.

Immunofluorescence Assay

Immunofluorescence studies were performed on 6- μ m optimal cutting temperature-embedded cryosections of adipose and heart tissue. The cryosections were fixed with 4% paraformaldehyde at room temperature for 15 minutes, washed, and blocked with PBS containing 5% BSA at room temperature for 2 hours. The sections were subsequently incubated overnight with the primary NaKtide antibody at 4°C , followed by washing and probing with a secondary antibody (Alexa Fluor 647 Red), and the slides were then mounted with 4',6-diamidino-2-phenylindole. Expression of GFP and NaKtide was assessed

using GFP and RFP filters, respectively, with a Nikon Eclipse 80i microscope equipped with a Nikon camera head DS-Fi1 (Nikon).

For immunofluorescence studies on the aortic sinus, after the blocking step, tissue sections were incubated overnight at 4°C with either a primary CD68 antibody or a primary α -SMA (α -smooth muscle actin) antibody. The following day, the slides were washed 3× with PBS and stained with a secondary antibody, Alexa Fluor 488 (Invitrogen), for 1 hour at room temperature. After washing 3× with PBS, the slides were visualized using an EVOS 7000 microscope (Thermo Scientific). Results were expressed as a percentage of cell marker–positive areas among the total aortic sinus area.

Statistical Analysis

Results are presented as means $\pm$ SE (SEM). All N values represent biological replicates. Normality was assessed using the Shapiro-Wilk test in combination with evaluation of scatter plots and QQ plots. For data that did not meet the normality assumption, nonparametric analyses were performed using the Kruskal-Wallis test with Dunn multiple comparisons. For normally distributed data, comparisons between the groups were performed using a 1-way and 2-way ANOVA, followed by Tukey test comparison for post hoc analysis. Statistical significance was assigned at $P < 0.05$. Data analyses were performed using GraphPad Prism 10.3 (GraphPad Software, San Diego, CA).

RESULTS

Adipo-NaKtide Expression Attenuated Atherosclerotic Plaque Formation in *Apoe*^{−/−} Mice Fed WD

Tissue specificity of the lentiviral construct expressing GFP-NaKtide under the control of the adiponectin promoter (Adipo-NaKtide) was assessed by immunofluorescence microscopy in VFAT from *Apoe*^{−/−} mice using an anti-NaKtide antibody. In mice injected with the Adipo-NaKtide construct, there was expression of GFP (green) and NaKtide (red) signals in VFAT (Figure 1A). In contrast, no NaKtide signal was detected in VFAT from mice injected with the control lentiviral vector (Adipo-GFP) or in heart tissues from either vector treatment group (Figure 1A; Figure S1A).

After 12 weeks of WD, atherosclerotic plaque formation in the aortic arch of male *Apoe*^{−/−} mice with or without Adipo-NaKtide treatment was evaluated by en face ORO staining. The WD-fed mice had lipid-rich atherosclerotic plaques covering 10.12 $\pm$ 1.70% of the total arch area, compared with 1.99 $\pm$ 1.00% in the CTR-fed mice. Adipo-NaKtide treatment reduced plaque area in the aortic arch by 67% in the WD-fed mice (Figure 1B). Similarly, cross-sectional plaque area at the level of the aortic sinus was reduced by 48% in WD-fed mice treated with Adipo-NaKtide lentivirus compared with the CTR-fed mice (14.37 $\pm$ 3.01% versus 7.55 $\pm$ 0.66%

plaque area; Figure 1C). Moreover, Adipo-NaKtide showed similar atheroprotective effects in female *Apoe*^{−/−} mice (Figure S1B). Because expression of the control lentiviral vector (Adipo-GFP) did not affect atherosclerotic plaque development, proinflammatory TNF- α expression, and plasma cholesterol levels in *Apoe*^{−/−} mice (Figure S1C through S1G), all subsequent experiments were conducted using the same 4 groups shown in Figure 1B.

Adipo-NaKtide Expression Reduced Macrophage Infiltration and Smooth Muscle Cell Content in Aortic Sinus Lesions of *Apoe*^{−/−} Mice Fed WD

Macrophages play a central role in the development and progression of atherosclerosis.²² Immunostaining for the macrophage marker CD68 revealed that Adipo-NaKtide expression reduced CD68-positive areas in the aortic sinus of WD-fed mice by 45% from 7.88 $\pm$ 0.39% to 4.37 $\pm$ 0.86% (Figure 2A). In addition to macrophage accumulation, smooth muscle cell proliferation contributes to plaque growth and stability.^{23,24} Immunostaining for α -SMA showed a 53% reduction in α -SMA–positive areas in the aortic sinus from 6.59 $\pm$ 0.69% to 3.11 $\pm$ 0.58% in the Adipo-NaKtide–treated WD-fed mice (Figure 2B).

Adipo-NaKtide Expression Suppressed WD-Induced Inflammation in Aorta of *Apoe*^{−/−} Mice Fed WD

In the aorta of male mice, mRNA levels of the proinflammatory genes *Tnf*, *Il6*, and *Ccl2* increased by $\approx$ 4-fold in WD-fed *Apoe*^{−/−} mice compared with those fed CTR (Figure 3A through 3C). Treatment with Adipo-NaKtide lentivirus downregulated expression of these genes in WD-fed *Apoe*^{−/−} mice by 39% to 44% compared with control mice. *Trem2* and *Adgre1* are markers of aortic lipid-laden foamy macrophages,²⁵ and *Pkm* encodes a glycolytic enzyme that is associated with abnormal glucose metabolism and chronic inflammation in macrophages from patients with coronary artery disease.²⁶ Expression of these genes was upregulated by WD (between 153 and 320% increase), and this was suppressed by treatment with Adipo-NaKtide lentivirus (Figure 3D through 3F). Moreover, female *Apoe*^{−/−} mice exhibited a response pattern to WD and Adipo-NaKtide indistinguishable from that observed in males in terms of the expression of proinflammatory markers and the foamy macrophage marker *Trem2* (Figure S2). Together, these data show that adipose tissue expression of NaKtide protected *Apoe*^{−/−} mice from diet-induced arterial inflammation and atherosclerosis in a sex-independent manner.

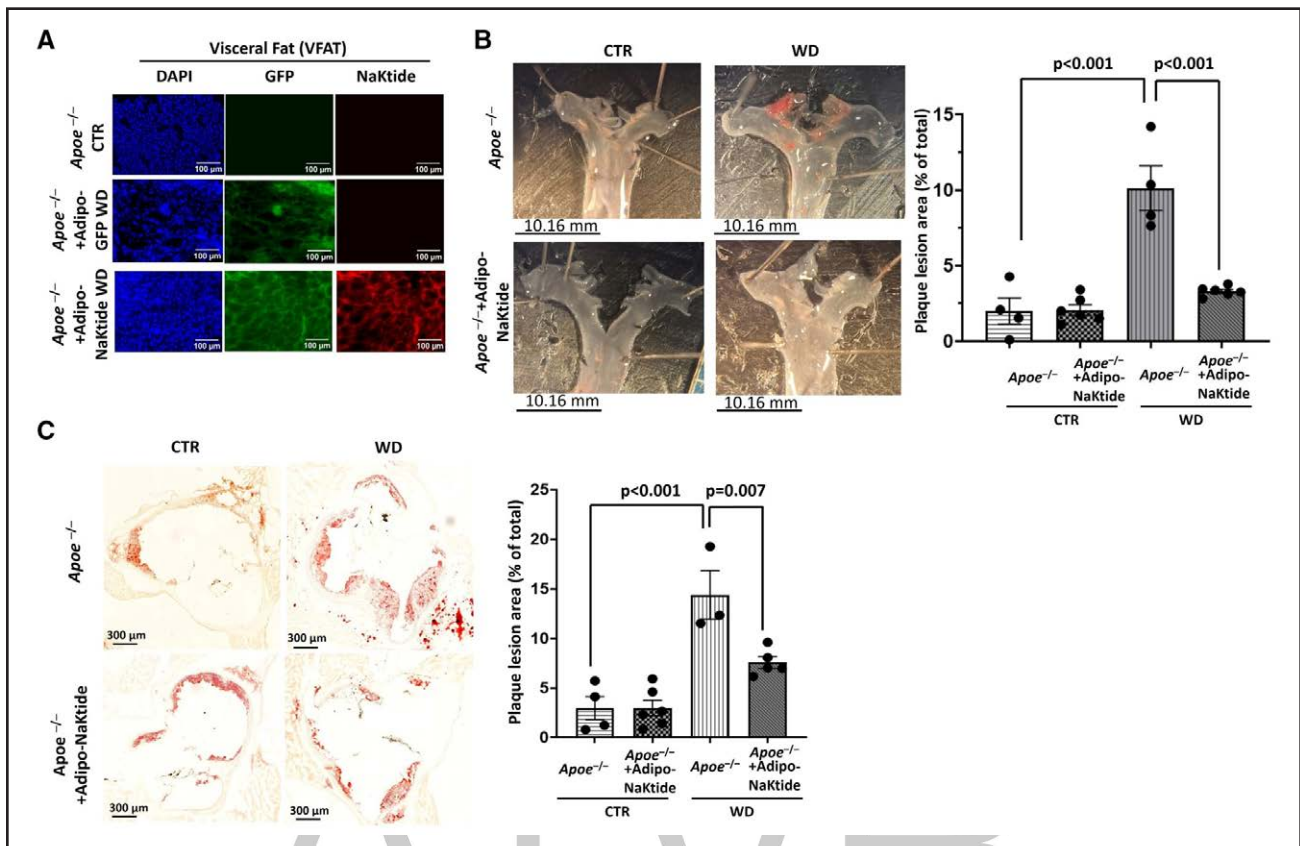


Figure 1. Adipocyte-specific expression of NaKtide attenuates atherosclerotic plaque formation in *Apoe*^{-/-} mice fed a Western diet (WD).

A, Immunofluorescence staining demonstrating expression of GFP (green fluorescent protein; green) and NaKtide (red) in visceral adipose tissue of mice transfected with adipocyte-specific NaKtide (Adipo-NaKtide; *n*=6 males per group). Scale bar: 100 μ m. **B, left**, Representative Oil Red O (ORO)-stained images of the aortic arch from *Apoe*^{-/-} mice fed a control diet (CTR) or WD for 12 weeks in the absence or presence of Adipo-NaKtide expression. **Right**, ORO-positive areas were quantified and are shown in the bar graph (*n*=4–6 males per group). Scale bar: 10.16 mm. **C, left**, Representative ORO-stained images of the aortic sinus from *Apoe*^{-/-} mice fed a CTR or WD for 12 weeks in the absence or presence of Adipo-NaKtide expression. **Right**, ORO-positive areas were quantified and are shown in the bar graph (*n*=3–6 males per group). Scale bar: 300 μ m. Values represent means $\pm$ SEM and were analyzed using 1-way ANOVA followed by the Tukey post hoc test. DAPI indicates 4',6-diamidino-2-phenylindole; and VFAT, visceral fat adipose tissue.

Adipo-NaKtide Expression Inhibited NKA Signaling and Reduced Oxidative Stress and Inflammation in Adipose Tissue of *Apoe*^{-/-} Mice Fed WD

WD feeding resulted in increased body weight, whereas adipo-NaKtide expression significantly attenuated WD-induced weight gain, with no effect observed under CTR-fed conditions (Figure S3A). In addition, analysis of food intake revealed no significant differences among the experimental groups (Figure S3B). Adipose tissue levels of protein carbonylation, an indicator of oxidative stress, were increased 3.3-fold in WD-fed *Apoe*^{-/-} mice compared with CTR-fed animals (0.85 $\pm$ 0.06 versus 0.26 $\pm$ 0.02 nmol/mg; Figure 4A). Adipo-NaKtide expression attenuated carbonylation levels to 0.45 $\pm$ 0.03 nmol/mg. Similarly, SFK phosphorylation was increased by 70 $\pm$ 22% by WD, and this increase was abolished by Adipo-NaKtide expression (Figure 4B). Furthermore, WD reduced AKT phosphorylation

by 24 $\pm$ 0.02%, consistent with impaired adipose insulin sensitivity, whereas Adipo-NaKtide expression rescued this decrease (Figure 4C). In addition, Adipo-NaKtide expression significantly attenuated WD-induced increased expression of genes that mediate adipocyte inflammation, including *Cd36*, *Il6*, and *Tnf* (Figure 4D through 4F). Adipo-NaKtide treatment also increased expression of *Ppargc1a*, a key regulator of mitochondrial biogenesis and oxidative stress²⁷ (Figure 4G). The effects on adipose inflammatory and metabolic markers were similarly observed in female *Apoe*^{-/-} mice (Figure S4).

Adipo-NaKtide Expression Improved Glucose Tolerance and Systemic Inflammation, and Reduced Expression of Circulatory Proinflammatory miRs in *Apoe*^{-/-} Mice Fed WD

Blood glucose levels in WD-fed *Apoe*^{-/-} mice were significantly elevated compared with CTR-fed controls

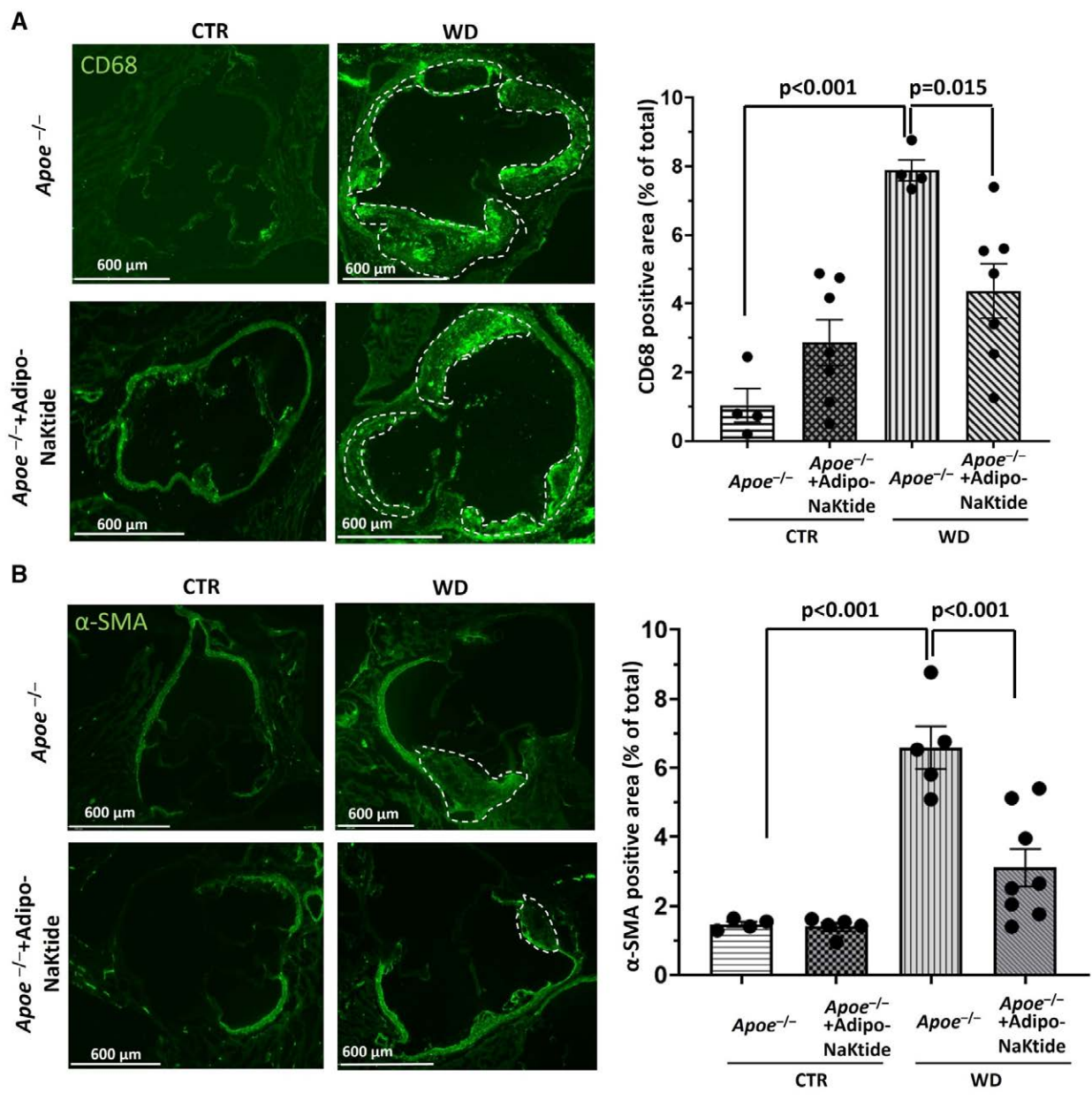


Figure 2. Adipocyte-specific expression of NaKtide reduces the cluster of differentiation 68 (CD68)- and α-SMA (α-smooth muscle actin)-positive area in *Apoe*^{-/-} mice fed a Western diet (WD).

A, left, Representative fluorescence images of the aortic sinus immunostained with anti-CD68 (green) from *Apoe*^{-/-} mice fed a control diet (CTR) or WD for 12 weeks in the absence or presence of adipocyte-specific NaKtide (Adipo-NaKtide) expression. **Right,** CD68-positive areas were quantified and are shown in the bar graph. Scale bar: 600 μm (n=4-7 males per group). **B, left,** Representative fluorescence images of the aortic sinus immunostained with anti-α-SMA (green) from *Apoe*^{-/-} mice fed a CTR or WD for 12 weeks in the absence or presence of Adipo-NaKtide expression. **Right,** α-SMA-positive areas were quantified and are shown in the bar graph. Scale bar: 600 μm (n=4-8 males per group). Values represent means±SEM and were analyzed using 1-way ANOVA followed by the Tukey post hoc test.

(Figure 5A), and Adipo-NaKtide treatment reduced these levels and improved glucose tolerance. In addition, the glucose level trend was equally observed in female *Apoe*^{-/-} mice (Figure S5A). Similarly, although WD increased plasma IL-6, TNF-α, and leptin levels to 297.46±9.32 pg/mL, 9.12±0.42 pg/mL, and 2140.08±154.89 pg/mL, respectively, Adipo-NaKtide

treatment lowered all the markers levels compared with WD-fed *Apoe*^{-/-} mice (129.15±15.10 pg/mL for IL-6, 7.36±0.40 pg/mL for TNF-α, and 1149.11±87.48 pg/mL for leptin; Figure 5B through 5D). A similar trend in IL-6 level was observed in female *Apoe*^{-/-} mice (Figure S5B). Furthermore, relative expression levels of circulating miR-27b-3p and miR-34a-3p, known mediators of

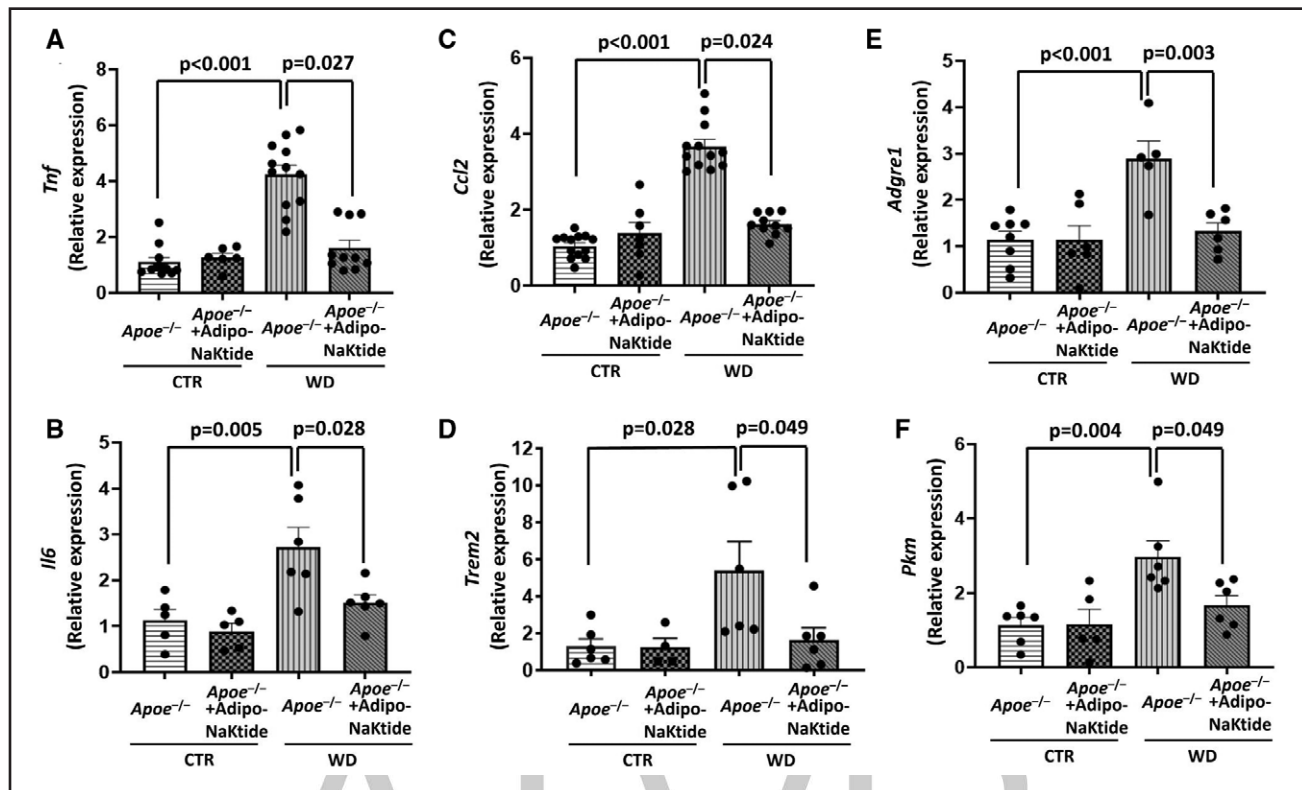


Figure 3. Adipocyte-specific expression of NaKtide improves inflammation and macrophage activation in the aortic tissue in *Apoe*^{-/-} mice fed a Western diet (WD).

Quantitative real-time polymerase chain reaction (qRT-PCR) analysis for (A) *Tnf* (n=6–12 males per group); (B) *Il6* (n=5–6 males per group); (C) *Ccl2* (n=7–13 males per group); (D) *Trem2*; (E) *Adgre1* (n=5–8 males per group); (F) *Pkm* (n=5–6 males per group). For all qRT-PCR analyses, GAPDH was used as a housekeeping gene. Values represent means±SEM and were analyzed using 1-way ANOVA followed by the Tukey post hoc test or Kruskal-Wallis test followed by the Dunn post hoc test. CTR indicates control diet.

atherosclerosis development,^{28,29} were downregulated by Adipo-NaKtide treatment in WD-fed *Apoe*^{-/-} mice (Figure 5E and 5F).

DISCUSSION

Atherosclerosis is increasingly recognized as a disease influenced by systemic metabolic and inflammatory signals beyond the vasculature.³⁰ Adipose tissue has emerged as a key modulator of vascular health through its endocrine and paracrine functions.³¹ Although adipocyte dysfunction is known to exacerbate oxidative stress and inflammation,^{6,32} the upstream signaling pathways mediating these effects remain incompletely understood. Here, we identify NKA signaling in adipocytes as a previously underappreciated contributor to atherogenesis.

Our study demonstrates that disruption of adipocyte NKA signaling via Adipo-NaKtide markedly attenuates atherosclerotic plaque development in WD-fed *Apoe*^{-/-} mice (Figure 1). These protective effects were accompanied by reduced macrophage accumulation in the aortic sinus (Figure 2), suggesting the importance of adipocyte-derived signals in shaping the vascular immune landscape. Notably, although the concept of adipose tissue

influencing vascular inflammation is not new, our work provides *in vivo* evidence that NKA signaling within adipocytes is a key upstream regulator of these proatherogenic processes.

Mechanistically, we showed that Adipo-NaKtide suppressed adipocyte oxidative stress and SFK activation (Figure 4), a hallmark of NKA signal transduction.¹⁰ This is consistent with our prior studies showing that NKA functions as a reactive oxygen species amplifier via a feed-forward mechanism involving SFK activation and carbonylation of the NKA $\alpha 1$ subunit.³³ The suppression of proinflammatory cytokine expression and restoration of *Ppargc1a* levels (Figure 4) further suggest that adipocyte NKA signaling governs a broader metabolic-inflammatory program, likely through mitochondrial dysfunction and redox imbalance.

A novel aspect of our findings is the potential requirement of NKA signaling for CD36-mediated adipocyte activation. Our previous work established that CD36 and NKA form a signaling complex in adipocytes and that oxLDL-induced CD36 signaling is blunted by pNaKtide.¹⁶ Here, Adipo-NaKtide reduced *Cd36* transcript levels and abrogated SFK activation *in vivo* (Figure 4), supporting a model in which NKA

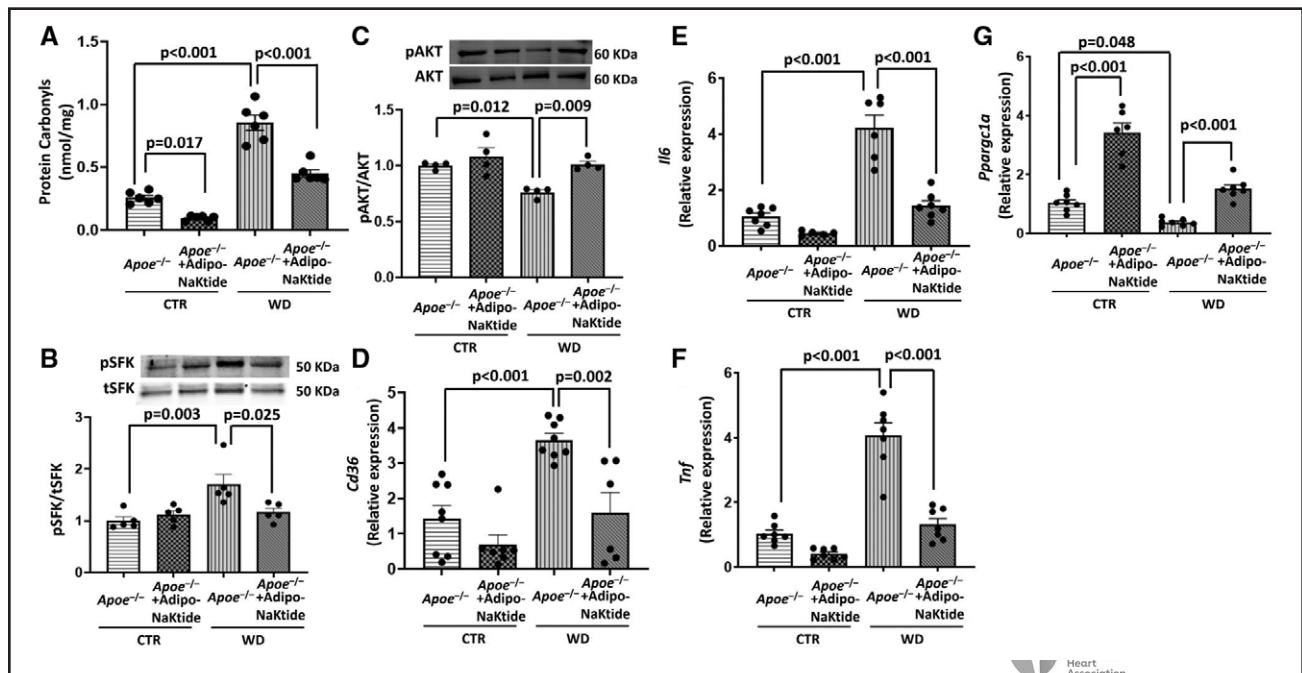


Figure 4. Adipocyte-specific expression of NaKtide improves oxidative stress, Na/K-ATPase signaling, and inflammation in visceral fat adipose tissue (VFAT) in *Apoe*^{-/-} mice fed a Western diet (WD).

A, Protein carbonylation ($n=6$ males per group). **B,** pSFK (phosphorylated Src-family kinase) immunoblot analysis with data shown as mean band densities normalized to total SFK (Src-family kinase; $n=5$ males per group). **C,** pAKT (phosphorylated protein kinase B) immunoblot analysis with data shown as mean band densities normalized to total AKT (protein kinase B; $n=4$ males per group). Quantitative real-time polymerase chain reaction (qRT-PCR) analysis for **(D)** *Cd36* ($n=6-8$ males per group); **(E)** *Il6* ($n=5-7$ males per group); **(F)** *Tnf* ($n=7$ males per group); **(G)** *Pparg1a* ($n=6-7$ males per group). For all qRT-PCR analyses, GAPDH was used as a housekeeping gene. Values represent means $\pm$ SEM and were analyzed using 1-way ANOVA followed by the Tukey post hoc test. CTR indicates control diet; and tSFK, total Src-family kinase.

is a key modulator or even a prerequisite for CD36-dependent pro-oxidant signaling in adipocytes. Given the established role of CD36 in lipid uptake,³⁴ mitochondrial metabolism,²⁰ and immune signaling,³⁵ this NKA-CD36 axis may serve as a central regulatory node linking lipid handling to inflammatory output in adipose tissue.

In addition, our data suggest that adipocyte NKA signaling influences systemic inflammation and glucose homeostasis (Figure 5). Adipo-NaKtide improved glucose tolerance and reduced circulating proinflammatory cytokines and miRs (miR-27b and miR-34a), both of which have been implicated in atherosclerosis.^{28,36} These findings reinforce the concept of adipose tissue as an active participant in vascular disease and highlight NKA signaling as a modifiable upstream target.

It is intriguing that suppression of adipocyte NKA signaling also led to a reduction in α -SMA-positive cells in the aortic sinus, most likely smooth muscle cells (Figure 2). Although the mechanisms remain unclear, this raises the possibility that adipocyte-derived factors regulated by NKA signaling can influence vascular smooth muscle phenotype and behavior in a non-cell-autonomous manner. Future studies are warranted to identify these factors and dissect their downstream effects.

We acknowledge several limitations. First, although NaKtide is undetectable in either the heart (Figure S1A) or the aortic sinus (not shown), it is not feasible for us to perform a comprehensive assessment of NaKtide expression across all vascular cell types or along the entire vascular tree. Therefore, although our data strongly support an adipose-driven mechanism, we cannot formally rule out a minor direct contribution from NaKtide expression in the vascular wall. A second limitation is the use of noninjected *Apoe*^{-/-} mice as controls rather than vector-injected controls, as lentiviral delivery may elicit innate immune responses. However, we show that adipocyte-specific control viral vector (Adipo-GFP) injection did not alter plaque burden, inflammatory markers, or plasma cholesterol (Figure S1B through S1G), supporting the conclusion that this experimental choice did not materially affect the overall findings. Third, the identity of the specific adipocyte-derived mediators remains to be determined. Fourth, although our model achieves adipocyte-restricted NKA inhibition, the long-term effects and safety of targeting this pathway systemically remain to be established. Finally, although our study provides strong evidence from a murine model, the relevance to human adipose tissue biology and atherosclerosis will require further validation.

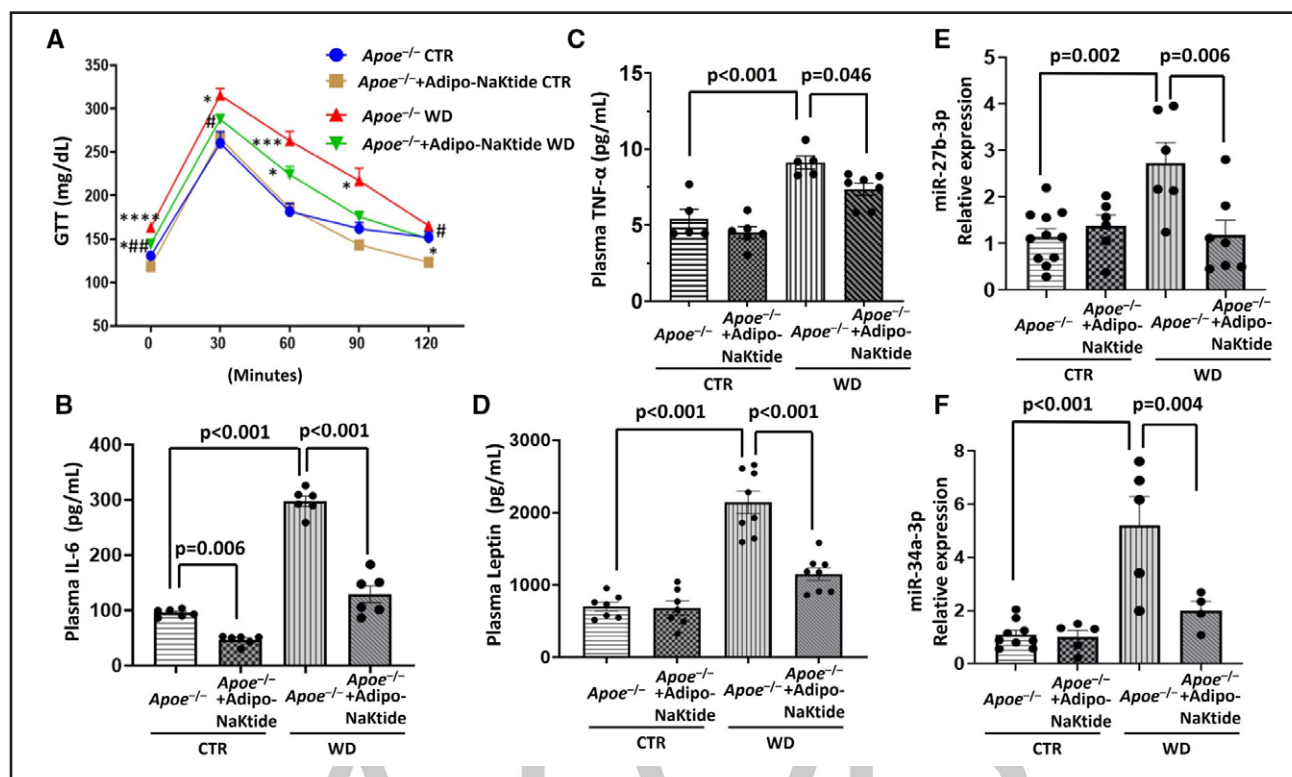


Figure 5. Adipocyte-specific expression of NaKtide improves metabolic profile, systemic inflammation, and circulating miRs in *Apoe*^{-/-} mice fed a Western diet (WD).

A, Glucose tolerance test. 0 minutes, **** $P < 0.001$ for *Apoe*^{-/-} WD, * $P = 0.018$, and ## $P = 0.004$ for *Apoe*^{-/-} + adipocyte-specific NaKtide (Adipo-NaKtide) WD; 30 minutes, * $P = 0.012$ for *Apoe*^{-/-} WD, # $P = 0.024$ for *Apoe*^{-/-} + Adipo-NaKtide WD; 60 minutes, *** $P < 0.001$ for *Apoe*^{-/-} WD, * $P = 0.020$ for *Apoe*^{-/-} + Adipo-NaKtide WD; 90 minutes, * $P = 0.019$ for *Apoe*^{-/-} WD; 120 minutes, * $P = 0.010$ for *Apoe*^{-/-} + Adipo-NaKtide control diet (CTR), # $P = 0.039$ for *Apoe*^{-/-} + Adipo-NaKtide WD ($n = 9-10$ males per group). Plasma concentration of **(B)** IL-6 (interleukin-6; $n = 6$ males per group); **(C)** TNF-α (tumor necrosis factor-α; $n = 5-7$ males per group); **(D)** leptin ($n = 7-8$ males per group). Quantitative real-time polymerase chain reaction (qRT-PCR) analysis for **(E)** miR-27b-3p ($n = 6-11$ males per group); **(F)** miR-34a-3p ($n = 4-9$ males per group). For all qRT-PCR analyses, U6 was used as an endogenous control. Values represent means ± SEM and were analyzed using 2-way ANOVA in **(A)** and 1-way ANOVA in **(B)** through **(F)**, followed by the Tukey post hoc test. GTT indicates glucose tolerance test. * P values indicate comparisons vs *Apoe*^{-/-} CTR, whereas # P values indicate comparisons vs *Apoe*^{-/-} WD.

In summary, we identify adipocyte NKA signaling as a novel driver of atherosclerosis, acting through oxidative stress amplification, modulation of inflammation, and potentially adipose-vascular communication (Figure 6). These findings establish a mechanistic framework for understanding how adipocytes contribute to vascular disease and open new avenues for exploring NKA signaling as a therapeutic target in metabolic and cardiovascular disorders.

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Author Contributions

B.S. Goncalves and Y. Wang performed the experiments, wrote the manuscript, and analyzed the data. S.S. Pillai, H.V. Lakhani, Q.Q. Luu, H. Triniac, and M.A. Chaudhry performed the experiments and analyzed the data. Z. Zheng assisted with the experimental design and data analysis. R.L. Silverstein, J.I. Shapiro, Y. Chen, and K. Sodhi acquired funding, designed the experiments, interpreted the data, and wrote and edited the manuscript. All authors contributed to the article and approved the submitted version.

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Disclosures

None.

Supplemental Material

Supplemental Methods
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ARTICLE INFORMATION

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